Permutations

A permutation of integers 1,2,…,n is called beautiful if there are no adjacent elements whose difference is 1.

Given n, construct a beautiful permutation if such a permutation exists.

Input

The only input line contains an integer n.

Output

Print a beautiful permutation of integers 1,2,…,n1. If there are several solutions, you may print any of them. If there are no solutions, print "NO SOLUTION".

Example 1 and 2

Input: 5	Input: 3
Output: 4 2 5 3 1	Output: NO SOLUTION

Approach

The idea is to construct a beautiful permutation by first listing all the even numbers up to n, followed by all the odd numbers up to n.
This arrangement guarantees that no two adjacent elements differ by exactly 1, because consecutive even numbers differ by at least 2, and the same holds for consecutive odd numbers.
The only exceptions occur when n = 2 or n = 3, for which no valid permutation exists.